

Enabling Omnichannel

Dealing with legacy in a digital world

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Where we are today?

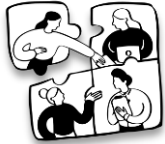


Customer Experience

Store inventory cannot be browsed online
Separate product inventory in channels limiting options
Check out, payments, delivery and support challenges

Business Operations

Separate order management in channels
Inefficient inventory management
Challenging support, reconciliations and reporting



Technology, Development and Operations

Lacks real-time integration and one source of truth
Poor mobile app experience
Inconsistent development processes resulting higher TTM



1 MM hits



Digital Platform



1000 users

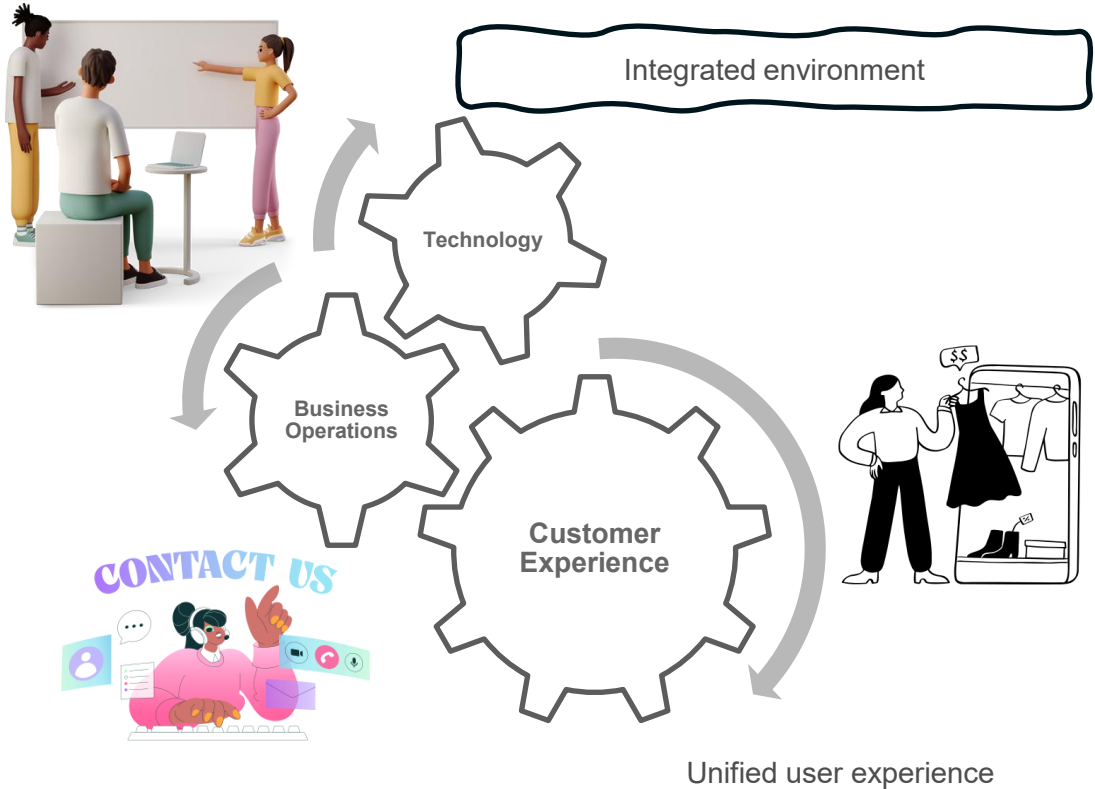
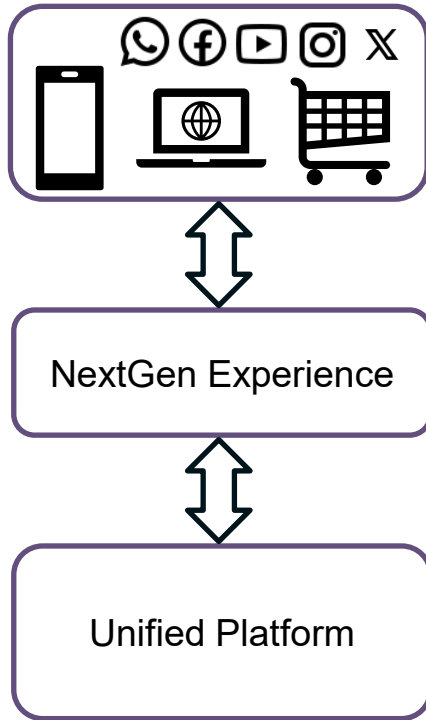


Legacy Platform

**But the good part is
we have a large loyal customer base
and quality products**

**with 30% market share and potential to grow up to 50% in next 3 years primarily driven by
increasing online shopping trend.**

Where we want to be?



What needs to be accomplished?

Customers should be able to

Browse inventory online and in stores.

Order online and pick-up from store

View, track online and offline orders and payments

Connect customer service or store sales representative

See product promotions on social media

Business

Customer Experience

- Consistent Web, Mobile and Store touchpoints
- Easy product discovery, ordering, checkout and tracking
- Personalization and Contextual
- Customer service
- Leverage social media
- Analytics

Unified Capabilities

- Product and Pricing Management
- Order Management
- Inventory
- Invoicing and Payments
- Customer Management
- Reporting and Business Intelligence

Technology

Architecture

- Common taxonomy
- User journey driven
- Real-time systems
- API and Event Driven
- Workflow based
- Elastically scale to handle internet traffic
- Harness power of data

Time to market

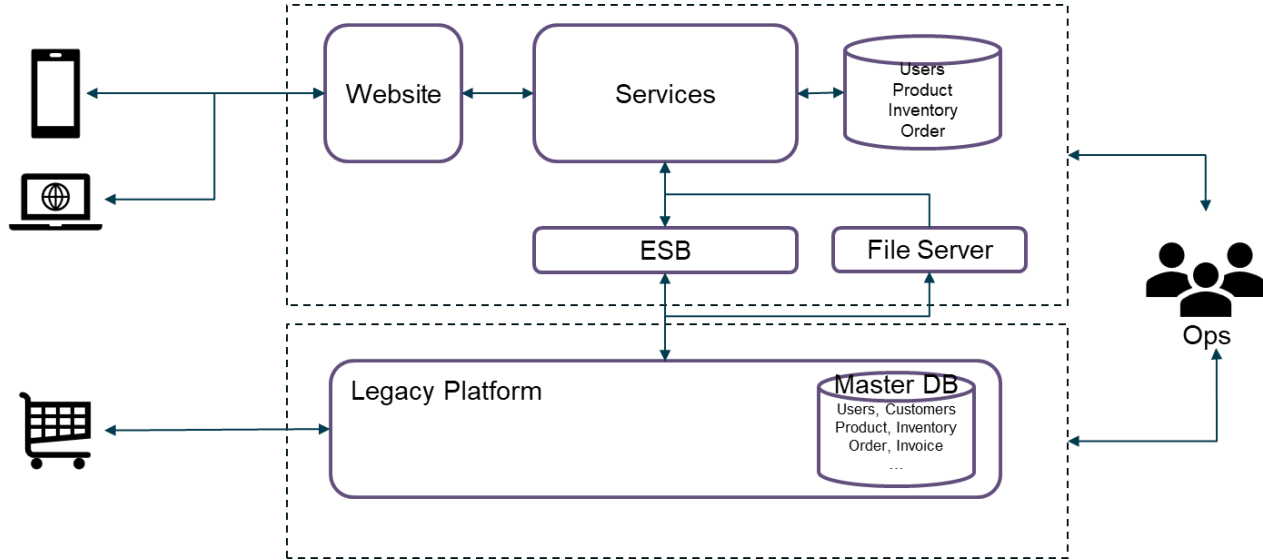
- Reimagined and re-engineered product and team construct for Core and Digital
- Align agile pods to customer journeys and SORs
- Customer centric and business value based prioritization
- Automation and metrics driven

While balancing Business and Technology drivers

Architecture

Current State (assumption)

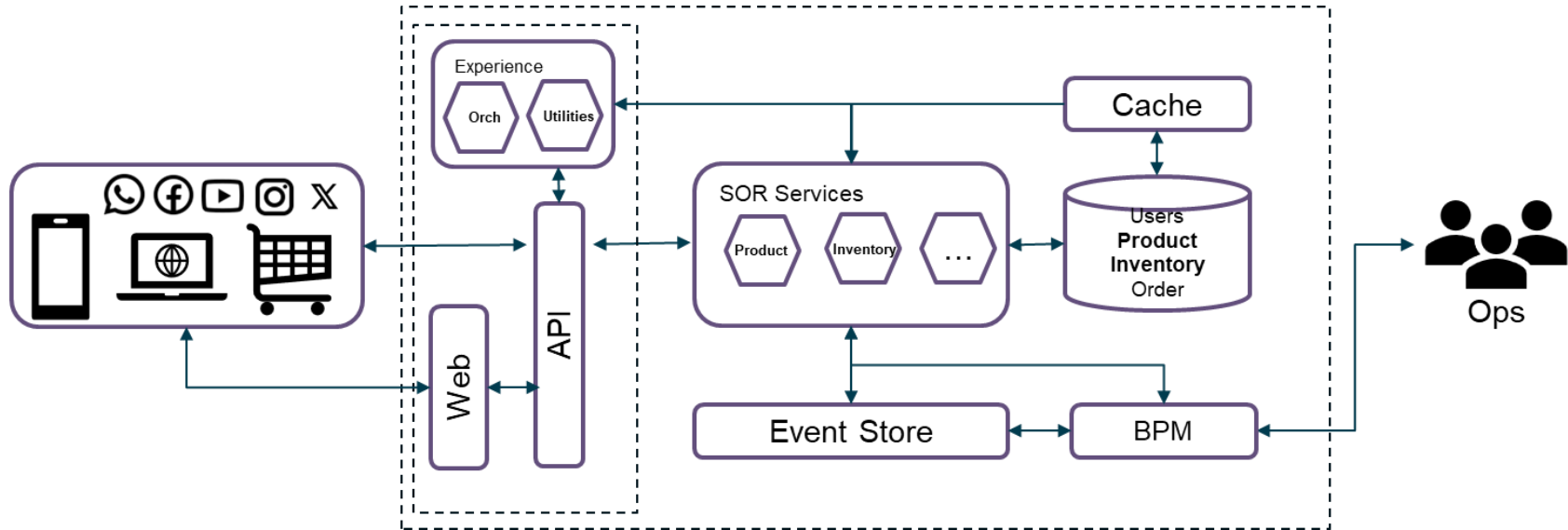
1. Redundant databases in Core and Digital
2. Product and Inventory data (delta snapshots) is shared via file as a copy
3. Orders and some other data is shared via an ESB.
4. Ops use different interfaces for Core and Legacy
5. Digital web interface uses services w/o API Gateway
6. Mobile app use web interface through a 3rd party “screen scrape” tool to extract Product and Inventory data.
7. Store users use legacy system for product, inventory, customer, order, payments etc.
8. Reporting is separate for Digital and Core and performed directly on databases.



- ❑ Currently there are around 1000 users across stores that use legacy system. Most transactions are orders and invoicing.
- ❑ Legacy application is hard to maintain and scale beyond a limit
- ❑ On Digital platform, currently there are ~1 million daily hits overall.
- ❑ On Digital ~85% traffic is for Product search

Desired State – Modernization is always desirable

1. In long term, deploy and scale the services depending on their varying throughput requirements. E.g. Product Search vs invoicing both have different scaling requirements. Take best out of infrastructure.
2. Separation of concerns such as creating best experience for end users vs managing system of records cleanly. Isolate with loose coupling and use domain driven design approach. However, possible reuse of existing digital services is recommended.
3. Use API, Event driven architecture to support seamless integration and business process orchestration.
4. Support agility and continuous delivery for faster TTM
5. Support clean team structure to manage dependencies so that there is less friction and higher time to market. Teams can be broadly organized as Experience layer, Core services layer and Operations and Internal services layer.

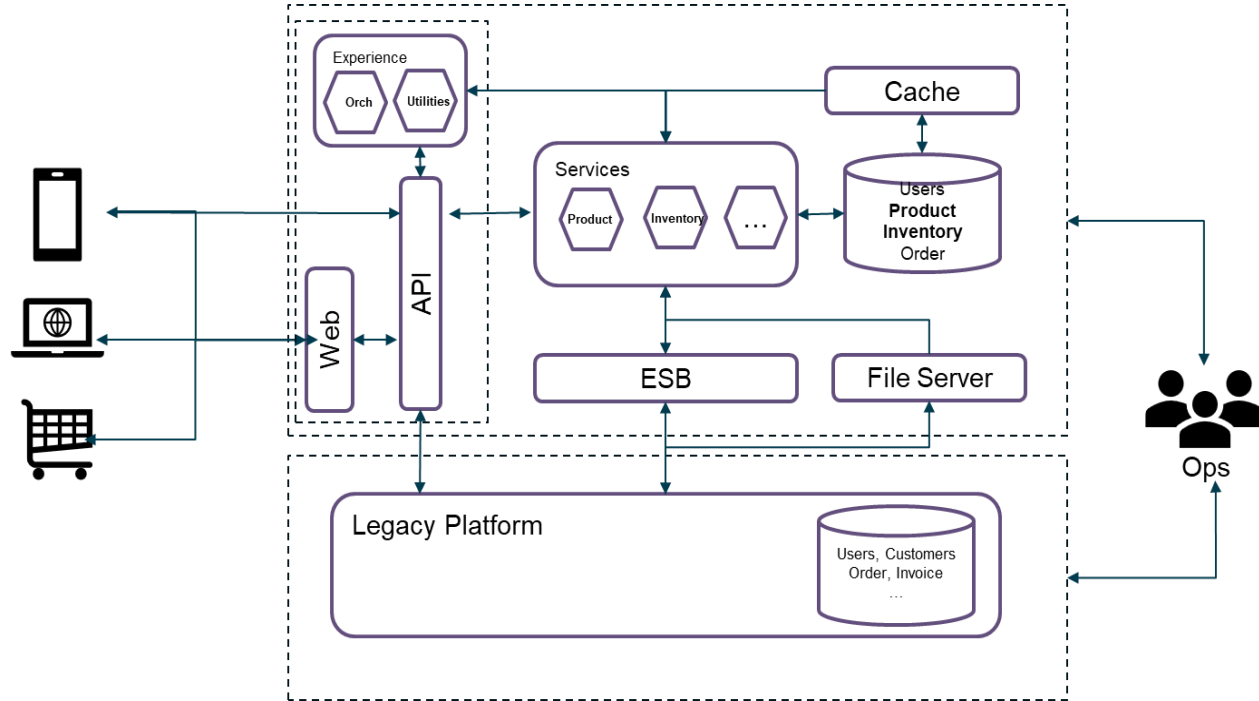


Transformation

Transition State(s)

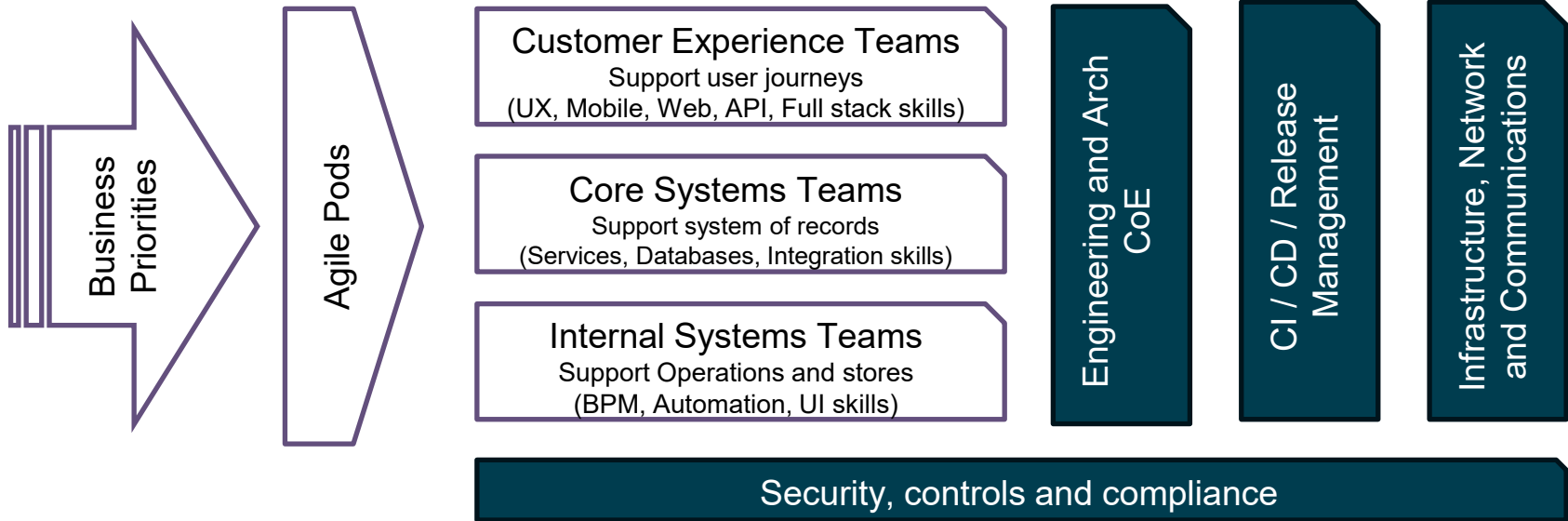
1. Capabilities needs to migrated towards a desired state in phased manner to avoid business disruption.
2. Enhance assets built for existing Digital
3. Use strangler pattern by introducing API layer to hide moving parts from the external world.
4. Introduce a layer to create desired user experience such as Search as you type or personalization etc
5. Move initial key services from Core to new platform that requires high throughput and unified view. Those are mostly Product and Inventory. Migrate the data.
6. Build essential tactical interfaces over legacy platform to support transition such as invoicing.
7. Minimize needs for database infrastructure and higher performance and throughput by introducing Caching for product and inventory at the minimum.
8. For other services, keep ESB and File sharing integration patterns.

Initiate transformation by organizing teams, redefining engineering processes and tools based on target state, unified and integrated delivery approach. Stabilize during transition.



Organize teams and processes

Teams to be organized to support agility, standardization and continuous improvement. This is not a headcount view but a capability view. Minimize the headcount and encourage teams to add value horizontally and vertically.



Roadmap activites

	H1	H2	H3	H4
People	Initial restructuring of teams keeping support for legacy inflight initiatives. Complete insourcing targets	Move more resources from Legacy to new platform. Reskilling or rehiring. Fine tune org structure Focus on SME maturity	Move most of legacy headcount to new platform keeping minimal support. Agile teams to focus on being self managed.	Decomm legacy teams No vendor resources beyond short term needs. Focus on ownership, talent on innovation mindset.
Process	Initial version of unified processes for requirement prioritization, product parties, development and release management. Handle impact on Ops and Store processes.	Redesign Ops and Store processes Fine tune processes established in H1. Increase release frequency. Establish key metrics across business process.	Enhance requirements, validation and development processes to be more efficient. Enhance continuous delivery. Adjust Ops process to handle more business. Metrics monitoring.	Decomm processes based on legacy systems. Monitor new process effectiveness and efficiencies.
Applications and Technology	Finalize open technology solutions Establish tooling such as Git, JIRA, Confluence, SeerviceNow, Monitoring etc. Build and migrate Order and Inventory from Core to new platform Build Web, Mobile and API initial versions. Build tactical interfaces for legacy.	Pilot launch of new / enhanced capabilities such as mobile App, website etc. Enhance the user experience. Tie user journeys and event driven integration. Stabilize pilot features and launch in production Establish monitoring and observability. Establish event store. Decomm screen scrape	Move other capabilities to new platform. Complete the data migration. Redevelop the reporting from one source. Stop using ESB. Move critical messaging to event based processes. Review and Optimize infrastructure	Decommission legacy system. Enhance experience layer to improve performance and scalability. Optimize infrastructure.

**Activities are indicative and will require formalization based constraints such as budget, timeline, technology etc.

Technology Choices

Capability	Choices to be considered
Mobile and Web interfaces	1. Flutter – Cross platform framework 2. React / NodeJS 3. Android and iOS native
API	Public Cloud API Gateway (Azure, AWS, GCP)
Services deployment	Docker, Kubernetes on cloud or on premise Or Serverless
Service Framework	Spring, NodeJS or others with reactive patterns
Databases	Oracle, PostGRESQL, SQLServer etc. However continue existing RDBMS of digital. RDBMS and NoSQL where applicable.
Event Store	Public cloud provided or On Premise such as Kafka
BPM / Workflow	Public cloud provided such as Azure LogicApps, AWS Step Functions and SWF
Others	Customer service tooling, chatbots, IVR etc
Observability	Grafana, Prometheus, / Kibana, Elastic Search etc. / AWS (CloudWatch, Xray), Azure App Insights

** These choices are a starting point. Further due diligence to be carried out. Need to leverage existing useful assets.

Assumptions and open items

- Management has informed about their aspirations. Current market share is about 30% and potential to grow 50% in next 3 years due to various internal and external factors
- Legacy platform already have core system of records such as Product, Customers, Inventory, Stores, Accounts etc.
- Legacy platform is a monolith application with an RDBMS as database. Support finite number of stores and internal users. Scaling the platform beyond a certain limit is challenging.
- Legacy platform is hosted on premise with infrastructure provisioned for stores and key transactions with digital.
- Digital platform is hosted on-premise as a web application.
- Digital and Legacy have evolved somewhat different taxonomy. Data metadata mapping and data lineage is becoming difficult.
- Operations use separate user interfaces for Digital and Core including reporting.
- Reporting for digital and core are different and reports are extracted directly from production databases.
- While in-sourcing mobile development, it is also expected to improve mobile user experience providing more native features

Key Open Items –

- Architecture and Deployment Strategy such as Cloud native vs On premise or hybrid
- Selection of platform such as Azure, AWS etc.
- Technology, framework choices for building blocks
- Agile, CI/CD, Testing, Engineering processes and tooling
- Estimations

Appendix

Option to keep legacy

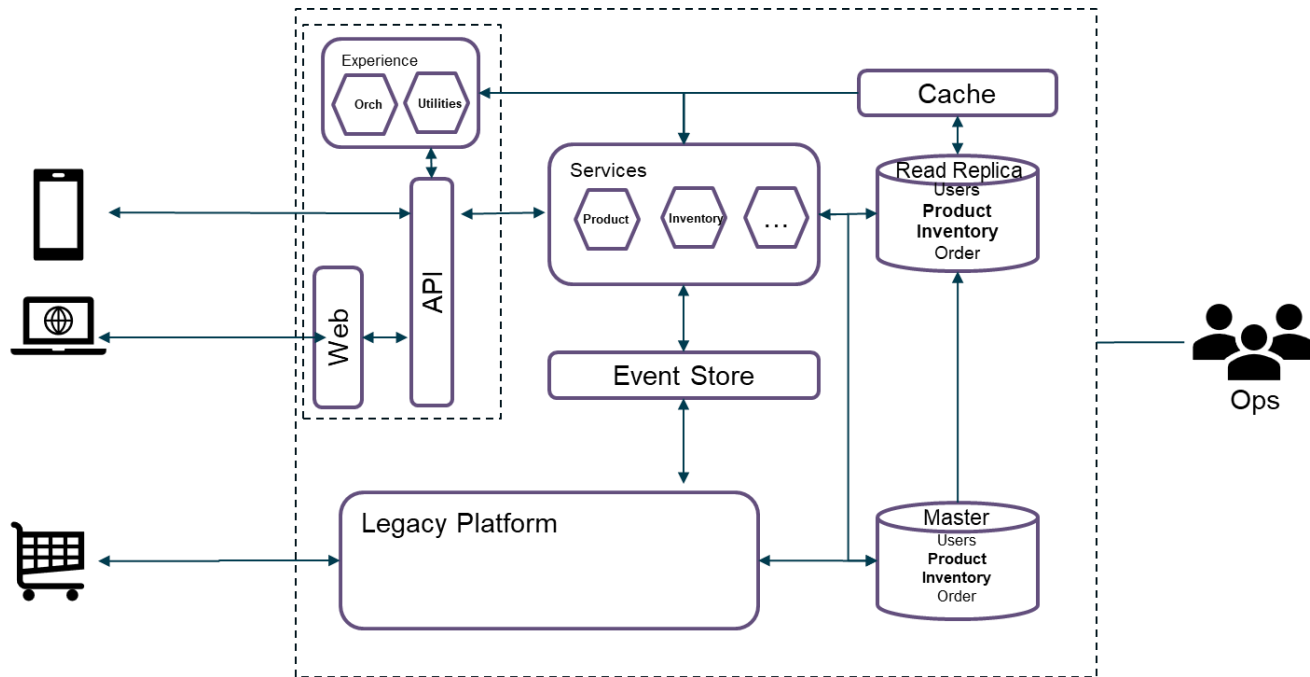
There are alternative patterns those are less invasive and leverage existing assets and investments.

This option provides a way to access common product, inventory, orders, logistics etc data.

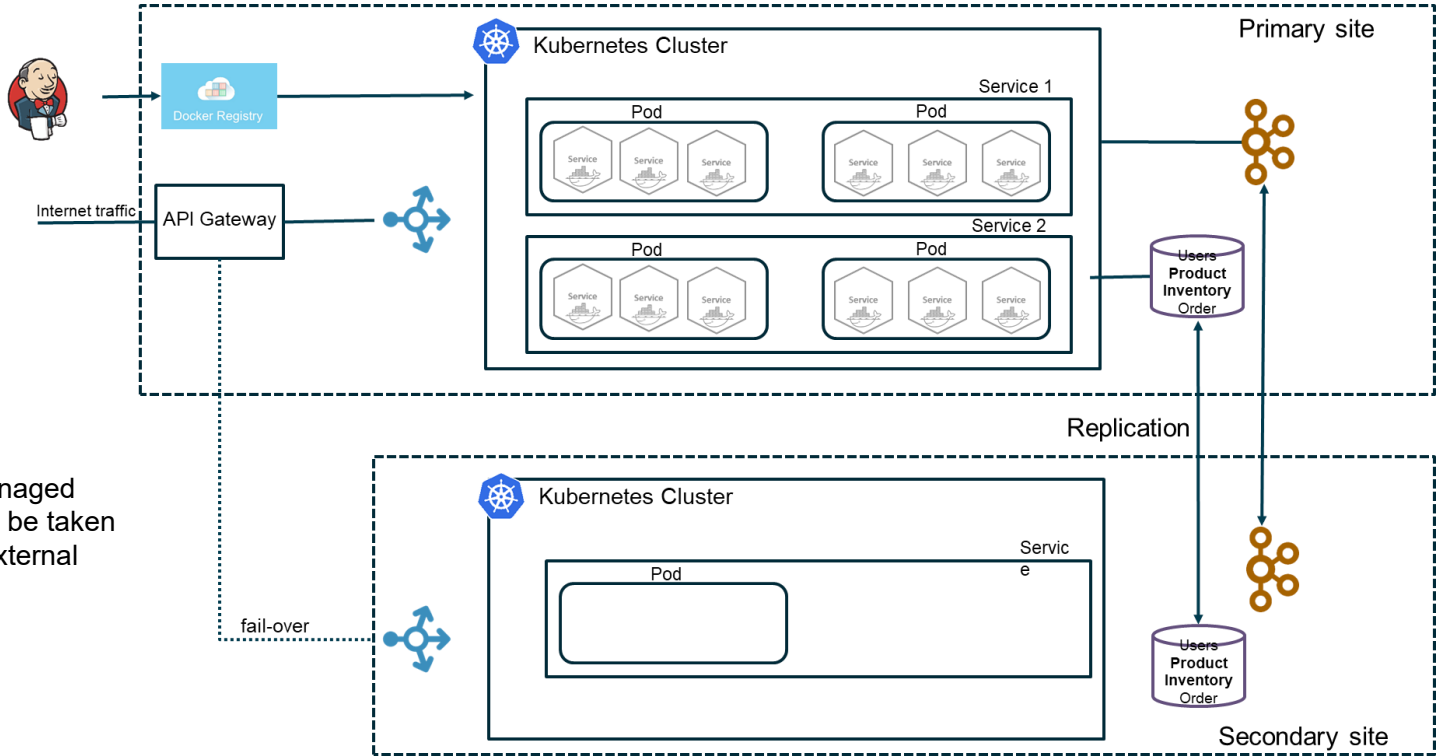
Read replica allows high throughput internet traffic, while keeping legacy with minimal change.

However, this leads to somewhat duplication of service / business logic in two places making to hard to manage cleanly. Deviates from DDD principles. Leads to tech debt towards modernization.

Hard to align agile team structure, skill sets, deployment and release management.



Service Deployment (Reference Architecture for desired state)



** Site resiliency of managed services is assumed to be taken care by the internal / external providers

Scenario

Scenario 1 - Dealing with legacy in a digital world: Enabling “Omnichannel”

A retail client wants to unify their online platform with their physical brick-and-mortar systems. Often referred to as Omnichannel in the retail space.

As has happened in many industries beyond retail, the foray into “digital” happened largely independently of the company’s core systems. In this particular client the “digital” group was a separate business unit, operating with an independent technology organization from the “mothership”. Integration was done at the legacy systems, through various means including a corporate ESB with pub-sub messaging and file sharing at its core. As examples, the digital group had its own copy of the product catalog, synced from the legacy systems via snapshot delta messages, but with additional capabilities around product artwork and other product metadata the digital group was driving faster on. Similarly, the digital inventory systems were unique to the digital channel but used message-based integration with core legacy systems. After an “internal” merger of the technical organizations, partially driven by the need to have a seamless experience across the physical and online channels, the organization needs to enable features like allowing a user to shop online but check out in the store, or browse store inventory from the web site.

In addition, the client has been using a 3rd party to “screen scrape” their online presence to generate their mobile site and now wants to bring the mobile development in house to enable greater flexibility in delivering across the physical and digital channels.

Note also that while the digital group “went agile” some 5+ years ago and does a fair bit of custom development, the core enterprise technology group has relied heavily on outsourcing from a number of different vendors over recent years and has only recently begun rebuilding its internal technical capabilities.

- What are the key things that will need to happen to enable seamless integration across channels from a customer perspective?
- What are the key technology practices that will enable rapid delivery in the new merged environment?